85

The common exponential function "natural exponential e^x " in series form can be written as

Answer

- $1+x-\frac{x^2}{2!}+\frac{x^3}{3!}-\ldots$

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- $x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots$
- $1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots$
- $1-x-\frac{x^2}{2!}-\frac{x^3}{3!}-$

Question No : 2 of 50

Marks: 1 (Time 1 Min)

A point x_0 of a differential equation that is not an ordinary point is said to be

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Answer

- (●) Non-singular point
- ( ) Singular point ✓
- ( ) Limit point
- ( ) None of them

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Question No : 3 of 50

Marks: 1 (Time 1 Min)

By applying the Operator method or systematic elimination on a system of linear homogeneous or linear non-homogeneous differential equations we always get a

Answer

- (•) Single linear differential equation
- ( ) Double linear differential equation
- ( ) Partial linear differential equation
- ( ) None of them

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Question No : 4 of 50

Marks: 1 (Time 1 Min)

Operator method is the method of the solution of a system of linear homogeneous or linear non-homogeneous differential equations which is based on the process of systematic elimination of the

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Answer

- Dependent variables
- Independent variable
- Choice variable
- None of them

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The solution of the linear first order differential equation $\frac{dy}{dx} - 2y = 0$ is

Answer

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Both (a) and (b)

None of them

Question No : 7 of 50

Marks: 1 (Time 1 Min)

For auxiliary equation of the D.E $x^2\frac{d^2y}{dx^2}-x\frac{dy}{dx}-10y=0$,substitute .....

Answer

- $y=x^m$
- $y=-x$
- $y=x$
- $y=x^2$

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Question No : 8 of 50

Marks: 1 (Time 1 Min)

The Differential Equation $(x^2+1)y'' + 2xy' + 6y = 0$ has singularity at

Answer

- $x = \pm 1$
- $x = \pm i$
- $x = \pm 2$
- $x = \pm 2i$

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Question No : 9 of 50

Marks: 1 (Time 1 Min)

Question No : 9 of 50

Marks: 1 (Time 1 Min)

If there exists a non-zero solution vector X of the system of linear differential equations such that $AX = \lambda X$. Then the number λ is said to be the

Answer

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- Random value
- None of them
- Matrix value
- Eigen value

Question No : 10 of 50

Marks: 1 (Time 1 Min)

Recurrence relation of the Differential equation $y'' + y = 0$ is

Answer

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- $c_{k+1} = -\frac{1}{k(k+1)} c_k$
- $c_{k+2} = -\frac{1}{(k+2)(k+1)} c_k$
- $c_{k+2} = \frac{1}{(k+2)(k+1)} c_k$

Question No : 11 of 50

Marks: 1 (Time 1 Min)

Singular points of the differential equation: $y'' + xy' + 2y = 0$

Answer

- is the set of all real numbers

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- is $x = 0$
- is $x = 1$
- does not exist

Question No : 12 of 50

Marks: 1 (Time 1 Min)

The differential operator form of the differential equation: $|a|y'' + by' - 2y = e^{|x|}$ for $a < 0$ is----.

Answer

- $(aD^2 + bD - 2)y = e^{|x|}$
- $(-aD^2 + bD - 2)y = e^{|x|}$
- $(aD^2 + bD - 2)y = e^{x}$
- $(-aD^2 + bD - 2)y = e^{-x}$

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Question No : 13 of 50

Marks: 1 (Time 1 Min)

If $E(t) = 0, R \neq 0$ (E(t) is the source voltage and R is the resistance) then electric vibration of the circuit is said to be

Answer

- Free damped oscillation
- Free un-damped oscillation
- Both damped and un-damped oscillation
- None of them

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Question No : 14 of 50

Marks: 1 (Time 1 Min)

The vectors $X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}$ and $X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$ are

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Answer

- (●) linearly dependent
- ( ) linearly independent

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Question No : 15 of 50

Marks: 1 (Time 1 Min)

The time interval between two successive maxima of $x(t) = Ae^{-\lambda t}Sin(\sqrt{\omega^2 - \lambda^2}\, t + \phi)$ is called

Answer

- Phase angle
- Amplitude
- Quasi period
- None of these

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Question No : 16 of 50

Marks: 1 (Time 1 Min)

In Cauchy-Euler Equation, to obtain the auxiliary equation we consider _____.

Answer

- $y = e^{mt}$
- $y = e^{2mt}$
- $y = e^{t}$
- $y = e^{m/t}$

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$\det(A - \lambda I) = 0$; this equation is called _____ .

Answer

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- eigen value
- characteristic equation
- eigen vector
- polynomial equation

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Question No : 18 of 50

Marks: 1 (Time 1 Min)

When the characteristic equation has repeated roots, it may _____ to find n linearly independent eigenvectors of the matrix.

Answer

- possible
- not possible
- totally possible
- always happen

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Question No : 19 of 50

Marks: 1 (Time 1 Min)

For finding the general solution of the non-homogeneous system of linear differential equations we need to find _____.

Answer

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- particular solution
- singular solution
- Both complementary and particular solution

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The non-homogeneous system of differential equations

$$\frac{dx}{dt} = 5x - 7y + 5\,Sint$$
$$\frac{dy}{dt} = 7x + 5y - 11\,Sint$$

Answer

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- $X' = \begin{pmatrix} 5 & -7 \\ 7 & -5 \end{pmatrix} X + \begin{pmatrix} 5 \\ -11 \end{pmatrix} Sint$
- $X' = \begin{pmatrix} 5 & -7 \\ 7 & 5 \end{pmatrix} X + \begin{pmatrix} 5 \\ -11 \end{pmatrix} Sint$

Let t denote any point in some interval denoted by I. Then the problem of solving the system of differential equations

$$\frac{dX}{dx} = A(t)X + F(t)$$

Subject to the _____ conditions

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Answer

- dirichlet
- neumann
- initial
- boundary

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Question No : 22 of 50

Marks: 1 (Time 1 Min)

By converting the equation $(x^2-4)^2y'' + (x-2)y' + y = 0$ into general form $y'' + P(x)y' + Q(x)y = 0$: p(x) is

Answer

- None of them

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- $\frac{1}{(x-2)(x+2)^2}$
- $\frac{1}{(x-2)^2(x+2)^2}$

Question No : 23 of 50

Marks: 1 (Time 1 Min)

Ir - regular singular point of the equation $(x^2-4)^2y'' + (x-2)y' + y = 0$ is

Answer

- $x = -2$
- $x = 2$
- None of them
- $x = -2, 2$

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Question No : 24 of 50

Marks: 1 (Time 1 Min)

The second order linear differential equation $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(x^2-49)=0$ is a Bessel equation of degree.

Answer

- None of them
- 1
- 2
- 3

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The general solution of the equation $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(x^2-\frac{1}{25})=0$ is

Answer

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$y=c_1J_{\frac{1}{4}}(x)+c_2J_{-\frac{1}{4}}(x)$

$y=c_1J_{\frac{1}{5}}(x)+c_2J_{-\frac{1}{5}}(x)$

Question No : 26 of 50

Marks: 1 (Time 1 Min)

If m and n are non negative integers and $p_n(x)$ is a Legendre's polynomial then

Answer

- $\int_{-1}^{1} P_m(x)P_n(x)dx = 0 \quad for\ m = n$

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- $\int_{-1}^{1} P_m(x)P_n(x)dx = 0 \quad for\ n < 0$

- $\int_{-1}^{1} P_m(x)P_n(x)dx = 0 \quad for\ m \neq n$

If $\frac{1}{2}\int_{-1}^{1} p_n(x)p_n(x)(2n+1)dx = 1$ then for $-1 < x < 1$ the Legendre's polynomial is said to

Answer

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- Orthogonal w.r.t weight function $w(x) = \frac{2n+1}{2}$
- Orthogonal w.r.t weight function $w(x) = 2n+1$

Question No : 28 of 50

Marks: 1 (Time 1 Min)

Transpose of a matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is

Answer

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- (•) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- ( ) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

Question No : 29 of 50

Marks: 1 (Time 1 Min)

If wroskian of the solution vectors X_1 and X_2 is zero, then vectors are ----.

Answer

- Parallel
- Linearly dependent
- Linearly Independent
- None of them

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Question No : 30 of 50

Marks: 1 (Time 1 Min)

If $\begin{vmatrix} 2-\lambda & -8 \\ 2 & -6-\lambda \end{vmatrix} = 0$ then it gives

Answer

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- $\lambda = -6$ of multiplicity 2
- $\lambda = 2$ of multiplicity 2
- $\lambda = 6$ of multiplicity 2
- $\lambda = -2$ of multiplicity 2

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The matrix $\begin{bmatrix} \sqrt{3} & 1 & 2 \\ 0 & \sqrt{3} & 1 \\ 0 & 0 & \sqrt{3} \end{bmatrix}$ has eigenvalue of multiplicity

Answer

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- 1
- 2
- 3
- $\sqrt{3}$

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Question No : 32 of 50

Marks: 1 (Time 1 Min)

The eigenvalues of homogeneous system $X' = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} X$, are

Answer

- $\lambda_1 = i, \lambda_2 = -i$
- $\lambda_1 = 1, \lambda_2 = -1$
- $\lambda_1 = 1, \lambda_2 = -i$
- $\lambda_1 = -1, \lambda_2 = -i$

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Question No : 33 of 50

Marks: 1 (Time 1 Min)

The matrix $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ has

Answer

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- Complex eigen value
- Repeated & real eigen value
- None of them
- Real and unequal value

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Question No : 34 of 50

Marks: 1 (Time 1 Min)

If a matrix $\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$, then eigenvalues are

Answer

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- 2,-1
- -1,-2
- 1,-2
- 2,1

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Go to Settin

Which of the following is correct differential operator form of the differential equation $2\frac{dx}{dt} - 5x + \frac{dy}{dt} = e^t$?

Answer

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- $(2D - 5)x + Dy = e^{2t}$
- $(2D - 5)x + Dy = e^{-t}$
- $(2D - 5)x + Dy = e^{t}$
- $(2D + 5)x + Dy = e^{t}$

If determinant $\begin{vmatrix} D-3 & 1 \\ -1 & D-1 \end{vmatrix} = 0$, then which of the following is true?

Answer

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- $D^2 - 4D + 1 = 0$
- $D^2 - 4D + 4 = 0$
- $D^2 - 4D + 3 = 0$
- None of these

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Marks: 1 (Time 1 Min)

What is the differential operator form of the differential equation $|a|y'' + by' + 2y = cosx$?

Answer

- $(bD + 2) = cosx$

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- $(-aD^2 + bD + 2)y = cosx$
- $(|a|D^2 + bD + 2)y = cosx$ ✓
- $(aD^2 + bD + 2)y = cosx$

Question No : 39 of 50

Marks: 1 (Time 1 Min)

What are the eigen values for the matrix $\begin{pmatrix} 1 & 5 \\ 0 & 7 \end{pmatrix}$?

Answer

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- {1,5}
- {1,0}
- {6,5}
- {1,7}

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Question No : 40 of 50

Marks: 1 (Time 1 Min)

What are the eigen values for the matrix $\begin{pmatrix} 0 & 1 \\ 6 & 7 \end{pmatrix}$?

Answer

- (•) {0,7}
- ( ) {1,6}
- ( ) {-1,-6}
- ( ) {7,5}

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Not confirm

Question No : 6 of 50

Marks: 1 (Time 1 Min)

Ordinary points of $(x^2-64)(x^2-36)y'' + xy' - y = 0$ are

Answer

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- 0,1
- 8,-8
- 6,-6
- None of others.

If $m_1 = 4+i$ and $m_2 = 4-i$ are the roots of Cauchy Euler differential equation then find the general solution.

Answer

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Background Color

As given that :

$m_1 = 4+i$

$m_2 = 4-i$

The general solution is :

Rewrite the given second order differential equation as a system in the normal form

$4y'' + y' + 2y = 0$

Answer

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Given equation :

$$4y'' + y' + 2y = 0$$

Now

dividing the equation by 4, *we get*

Activate

Find the Wronskian of the solutions $X_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-8t}$ and $X_2 = \begin{pmatrix} -2 \\ 3 \end{pmatrix} e^{-4t}$ of a system.

Answer

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Write the system as :

$$X_1 = \begin{bmatrix} -e^{-8t} \\ e^{-8t} \end{bmatrix} \quad \text{and} \quad X_2 = \begin{bmatrix} -2e^{-4t} \\ 3e^{-4t} \end{bmatrix}$$

using a Wronskian method

Write the system as :

$$X_1 = \begin{bmatrix} -e^{-8t} \\ e^{-8t} \end{bmatrix} \quad and \quad X_2 = \begin{bmatrix} -2e^{-4t} \\ 3e^{-4t} \end{bmatrix}$$

using Wronskian method

$$W(X_1, X_2) = \begin{vmatrix} -e^{-8t} & -2e^{-4t} \\ e^{-8t} & 3e^{-4t} \end{vmatrix}$$

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Now

$$= 3e^{-8t-4t} + 2e^{-8t-4t}$$

$$= 3e^{-12t} + 2e^{-12t}$$

$$= 5e^{-12t}$$

Question No : 44 of 50

Marks: 3 (Time 9 Min)

State and explain the three forces acting on a mass attached with a spring according to Damped Motion?

Answer

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Therefore, there are three forces acting on the mass, namely:

a) Weight mg of the body

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b) Restoring force $-k(s+x)$

c) Damping force $-\beta\left(\frac{dx}{dt}\right)$

Question No : 45 of 50

Marks: 3 (Time 6 Min)

Why we take $c_0 \neq 0$ in Frobenius method?

Answer

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To find coefficient of the power series and values of c1,c2,c3...cn
We put the c0=0

Since nothing is gained by taking $c_0 = 0$, we must then have

$r(3r-2)=0$ [called the indicial equation and its roots $r = \frac{2}{3}, 0$ are called indicial roots or exponents of the singularity.]

Find the Eigen values of the given matrix

$$A = \begin{bmatrix} 1 & 4 \\ -4 & -7 \end{bmatrix}.$$

Answer

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for eigenvalues

$$A = \begin{bmatrix} 1-\lambda & 4 \\ -4 & -7-\lambda \end{bmatrix}$$

now taking det erimnate we have :

for eigenvalues :

$$A = \begin{bmatrix} 1-\lambda & 4 \\ -4 & -7-\lambda \end{bmatrix}$$

now taking det *erimnate we have :*

$$= (1-\lambda)(-7-\lambda)+16$$

$$= -7-\lambda+7\lambda+\lambda^2+16$$

$$= \lambda^2+6\lambda+9$$

$$= (\lambda+3)^2$$

$$(\lambda+3)^2 = 0$$

$$\lambda+3 = 0$$

$$\lambda = -3$$

Show that vectors $X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-6t}$ and $X_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-2t}$ are linearly dependent.

Answer

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Write the system as :

$$X_1 = \begin{bmatrix} e^{-6t} \\ -e^{-6t} \end{bmatrix} \quad \text{and} \quad X_2 = \begin{bmatrix} -e^{-2t} \\ e^{-2t} \end{bmatrix}$$

using a Wronskian method

Activate

Write the system as :

$$X_1 = \begin{bmatrix} e^{-6t} \\ -e^{-6t} \end{bmatrix} \quad \text{and} \quad X_2 = \begin{bmatrix} -e^{-2t} \\ e^{-2t} \end{bmatrix}$$

using Wronskian method

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$$W(X_1, X_2) = \begin{vmatrix} e^{-6t} & -e^{-2t} \\ -e^{-6t} & e^{-2t} \end{vmatrix}$$

Now

$$= e^{-6t-2t} - e^{-6t-2t}$$

$$= e^{-8t} - e^{-8t}$$

$$= 0$$

This is linearly independant.

Question No : 48 of 50

Marks: 5 (Time 10 Min)

Write down the procedure of solution of the system of differential equations by "Operator Method".

Answer

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Step 1. Write in operator from.
Step 2. Eliminate the variable of x
Step 3. Make the auxiliary equation of step 2
Step 4. Eliminate the variable of y
Step 5. Make the auxiliary equation of step 4
Step 6. Put the x(t) and y(t) in the any given the equations
Step 7. Write the general solution.

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Question No : 49 of 50

Marks: 8 (Time 10 Min)

Find the first four terms of a power series in x for the function $e^x \sin x$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

Answer

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Question No : 50 of 50 Marks: 8 (Time 10 Min)

Solve the following Cauchy-Euler differential equation by suitable substitution,

$$x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+4y=0$$

Answer

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Normal form.

$4y'' + y' + 2y = 0$

Sol.

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Dividing the equation by 4, we get

$$y'' + \frac{1}{4}y' + \frac{1}{2}y = 0$$

Now variable

$y = x_1$, $y_2 = x_2$

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$x_1' = y_1' = x_2$

$x_2' = y_2''$

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Putting in equation

$$x_2' + \frac{1}{4}x_1' + \frac{1}{2}x_1 = 0$$

Given that:

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + 4y = 0$$

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Write it in operator form:

$$(x^2D^2 + xD + 4)y = 0$$

Substitution $x = e^t$ $t = \ln x$

$xD = A$, $x^2D = A^2 - A$

$P_4(6in_2)$

$(D^2 - D^2 + D + 4)y = 0$

$(D^2 + 4)y = 0$

$\frac{d^2y}{dt^2} + 4y = 0$

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make auxiliary:

$m^2 + 4 = 0$

$m^2 = -4$

$\Rightarrow m = \pm 2i$, $\alpha = 0$, $\beta = 2$

The root are complex

The General sol is:

$$y = e^{\alpha t}(C_1 \cos \beta t + C_2 \sin \beta t)$$

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$$y = e^{0t}(C_1 \cos 2t + C_2 \sin 2t)$$